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Belgaum, Karnataka- 590014



Third Year Project Report On “CAREER PATH FORECASTING SYSTEM”

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BACHELOR OF ENGINEERING
In
COMPUTER SCIENCE AND ENGINEERING
DATA SCIENCE

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Certificate

This is to certify that the Project Work entitled “**Career Path Forecasting System**” is a Bonafide work carried out by **Abhishek G (1DS23CD003)**, **Amarnath V (1DS23CD005)**, **K Yuvraj Sundaram (1DS23CD024)**, **Sambhram Naregal (1DS23CD045)** in partial fulfilment for Software Engineering subject for the V semester of Bachelor of Engineering in CSE (Data Science) of the Visvesvaraya Technological University, Belgaum during the year 2025-2026. The Project report has been approved as it satisfies the academics prescribed for the Bachelor of Engineering degree.

Signature of the Guide		Signature of HOD

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Abstract

Traditional academic evaluation, reliant solely on GPA, is an incomplete predictor of student potential in the modern career landscape, failing to capture the significant value of internships, certifications, and co-curricular achievements. The Career Path Forecasting System addresses this gap as a holistic, data-driven web application designed for personalized career guidance. Its core innovation is the "Academic Performance Indicator (API) Score," a novel composite metric that intelligently weighs a student's complete profile across all these areas. Built with Python, Pandas, and Scikit-learn, the system employs a Random Forest algorithm to analyze this holistic profile and predict the most suitable career path—Placement, Higher Studies, or Startup. This prediction is delivered via an interactive Streamlit dashboard, empowering students by replacing ambiguity with actionable, data-driven insights to help them make confident, well-informed career decisions.

CHAPTER 1

INTRODUCTION

1.1. Introduction

For decades, the global education ecosystem has been dominated by a single performance parameter: academic grades. In most higher education institutions, a Grade Point Average (GPA) or cumulative percentage is the primary, and often sole, measure of a student's capability and potential. However, as industries rapidly evolve, driven by technological innovation and global competition, this one-dimensional metric is increasingly found to be an incomplete and often misleading predictor of future success.

The modern student partakes in a rich and varied ecosystem of learning and development. This includes practical industry internships, specialized online certifications from global platforms, participation in high-level co-curricular competitions like hackathons, and research publications. These activities are critical to their overall development, fostering practical skills, problem-solving abilities, and adaptability—qualities that are highly sought by employers but are completely overlooked by traditional grading systems. Employers and graduate school admissions committees are progressively recognizing that exam scores alone are insufficient. A student with practical internship experience, specialized technical certifications, and a strong problem-solving portfolio may be far more industry-ready than a student with a perfect GPA but no applied skills.

1.2. Problem Statement

The core problem this project addresses is the significant and growing **disconnect between traditional academic evaluation and the multi-faceted requirements of the modern career landscape**.

- **For Students:** This gap creates immense uncertainty and anxiety. Students are unable to gauge their true standing and are often unsure of which career path—be it placements, further studies, or entrepreneurship—best aligns with their complete profile.
- **For Institutions:** Universities struggle to effectively guide their students and measure the "employability" of their cohorts beyond simple placement percentages.
- **For Employers:** Recruiters must sift through thousands of applications, relying on an unreliable primary filter (CGPA), which may cause them to overlook high-potential candidates who excel in non-academic areas.

The system addresses this problem by providing a holistic, data-driven tool for personalized career guidance that looks at the *whole* student, not just their transcript.

1.3. Purpose

The main objective of the "Career Path Forecasting" project is to **empower students with a personalized, data-driven, and holistic approach to selecting the right career path.**

The specific objectives are:

- To design and implement a novel "Academic Performance Indicator (API) Score" that aggregates and intelligently weighs academics, internships, certifications, and co-curriculars.
- To assess and quantify a student's strengths by analyzing their complete profile, identifying areas of excellence and potential improvement.
- To develop and train a highly accurate machine learning model (Random Forest) capable of predicting the most suitable career path.
- To provide students with tailored, actionable recommendations (Placement, Higher Studies, or Startup) based on their individual, holistic profile.
- To improve student decision-making by replacing ambiguity and intuition with data-driven insights, helping them make well-informed, confident decisions.

1.4. Scope

The scope of this project is to design, develop, and test a standalone web application.

- **Users:** The system is designed for use by **students** (from sophomore to senior year), **recent graduates**, and potentially **career changers** seeking personalized guidance.
- **Functionality:** The project's scope covers the entire lifecycle, from user data input (in a structured format) to the final prediction output. This includes data validation, the calculation of the composite API score, and the execution of the machine learning model.
- **Technology:** The scope is defined by the application of **supervised machine learning**, specifically the **Random Forest** algorithm, for the prediction engine. The user-facing component is an interactive web dashboard built with the **Streamlit** library.
- **Boundaries:** The system does *not* interface with live university databases. Data input is manual by the user. It does *not* provide specific job-matching (e.g., "Apply to Company X") but rather the strategic path (e.g., "Placement").

1.5 Process Model

In the development of the *Career Path Forecasting System (CPFS)*, the **Incremental Process Model** was adopted. This model was chosen because it supports **progressive development and delivery of the system in multiple increments**, rather than attempting to build the entire product at once. Each increment adds new functionality to the existing system until the final, complete software is delivered.

The incremental approach divides the project into several smaller, manageable modules, with each increment going through all the fundamental phases of software engineering — **Requirement Analysis, Design, Implementation, and Testing**. This ensures that every part of the system is developed, tested, and validated before moving on to the next increment, thereby minimizing risks and improving flexibility.

Key Characteristics of the Incremental Model

1. The total system is broken into smaller builds (increments), each adding more functionality.
2. Each increment passes through all SDLC phases (analysis, design, code, test).
3. Feedback from each increment is used to refine the next phase of development.
4. Early increments deliver a basic working version of the product that can be demonstrated to users.
5. Later increments enhance the system with advanced modules and features.

Advantages of Using the Incremental Model

- **Early delivery:** A basic version of the system is available in the first few weeks, allowing early testing and feedback.
- **Flexibility:** New or refined requirements can be added in later increments.
- **Risk reduction:** Issues are detected early since each increment is tested independently.
- **Customer feedback integration:** Continuous user input helps align the product with actual needs.
- **Resource efficiency:** Development and testing efforts can be distributed incrementally over time.

Application to the CPFS Project

The *Career Path Forecasting System* was developed in **three major increments**, each building upon the previous one:

- **Increment 1 – Core User Interface and Data Forms:**
This stage involved requirement analysis, UI design, and creation of student input forms for entering academic, co-curricular, internship, and certification details. The output of this increment was a fully functional **data entry interface** connected to a local database.
- **Increment 2 – Machine Learning Model Integration:**
In this phase, the system was enhanced by integrating the **Random Forest** and **Decision Tree** models for API score computation and career prediction. Data preprocessing, normalization, and ML logic were implemented and validated. The deliverable was a **working prediction engine** capable of generating API scores and recommending career paths.

- **Increment 3 – Visualization, Dashboard, and Reporting:**

The final increment added an **interactive dashboard**, performance charts, and the ability to **download PDF reports**. System-wide testing and refinement were conducted to ensure a stable, user-friendly, and accurate final product.

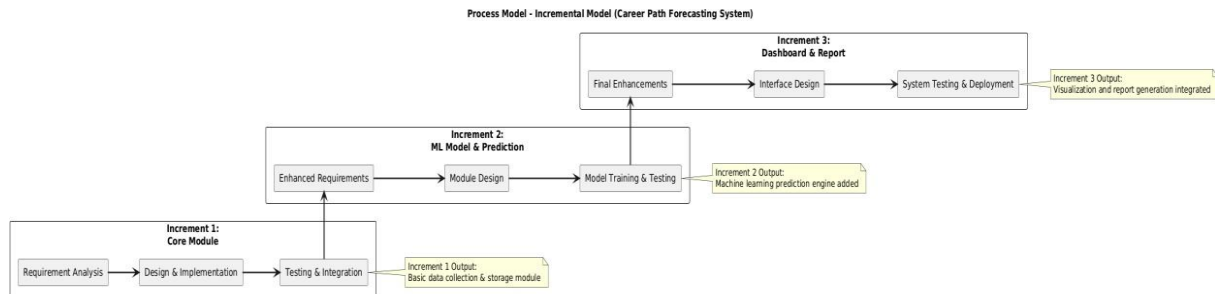


Figure 1.1: Process Model

Incremental Process Model showing the four development phases with their respective timelines and deliverables.

1.6. ER Diagram

The Entity-Relationship (ER) diagram for the system's database is centered around the Student entity, capturing all data points necessary for the API calculation and prediction.

- **Entities:** Student, Internship, Certification, Cocurricular, Prediction, Career Path.
- **Relationships:** A Student (1) *has* (M) Internship, Certification, and Cocurricular records. A Student (1) *receives* (M) Prediction results. A Prediction (M) *is mapped to* (1) Career Path.

Entity-Relationship Diagram for Career Path Forecasting System

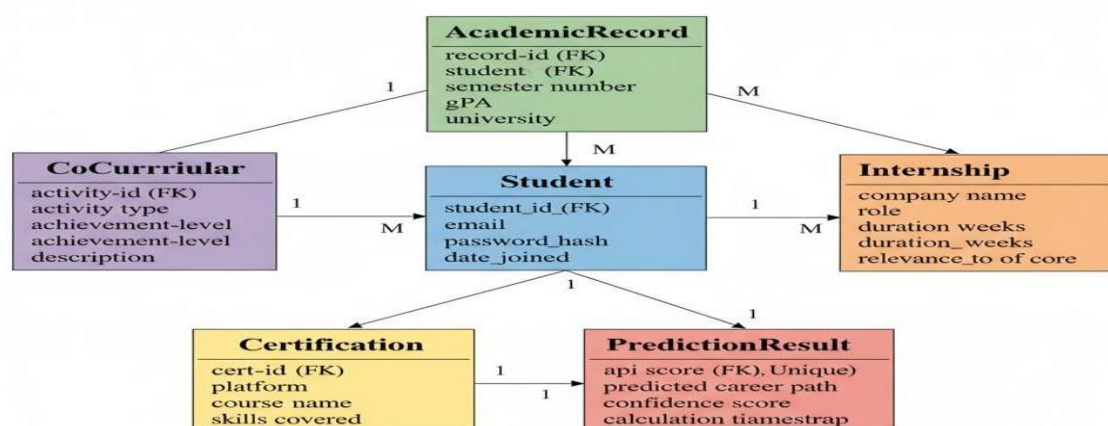


Figure 1.2:

Entity-Relationship Diagram illustrating the database structure and relationships between system entities.

CHAPTER 2

SOFTWARE REQUIREMENT SPECIFICATIONS

2.1. Overall Description

2.1.1. Product Perspective

The "Career Path Forecasting" system is a **standalone, interactive web application**. It is designed to be a self-contained product, accessed by end-users (students) via a standard web browser.

- **Backend:** The core logic is built in **Python 3.x**. It leverages the **Pandas** library for data manipulation and API score calculation, and **Scikit-learn** for executing the pre-trained machine learning model.
- **Frontend:** The user interface is a user-friendly dashboard built with **Streamlit**.
- **Interface:** The system's primary interface is with the human user. It does not require real-time integration with other external systems.

2.1.2. System Features

1. **User Authentication:** Secure user registration, login, and session management.
2. **Student Profile Management:** A comprehensive, multi-part form for students to input, view, and update their data (Academics, Internships, Certifications, Co-curriculars).
3. **API Score Calculation:** An internal backend module that processes the complete user profile and applies a weighted algorithm to calculate the composite API score.
4. **Career Path Prediction:** The core feature where the user's profile vector is fed into the trained Random Forest model to generate a prediction.
5. **Results Visualization Dashboard:** An interactive and easy-to-understand results page that displays the API score and the recommended career path.

2.1.3. User Characteristics

The primary users are **students** (sophomore to senior), **recent graduates**, and **career changers**.

- **Persona 1: "The Confused Sophomore"** - Has decent grades but no practical experience. Uses the system to understand their "API Score" and see what factors they are missing.
- **Persona 2: "The Ambitious Junior"** - Has a strong profile. Is at a crossroads: "Placements or Master's?" Uses the system to get a data-driven "second opinion."
- **Assumed Expertise:** Users are expected to have basic computer and web literacy. **No technical or data science expertise is required.**

2.1.4. Functional Requirements

The functional requirements define the specific behaviors and functions of the system.

Req. ID	Category	Requirement Description
FR-1.1	User Mgmt	The system shall allow a new user to register with a name, unique email, and password.
FR-1.2	User Mgmt	The system shall hash and salt the user's password before storage.
FR-1.3	User Mgmt	The system shall allow a registered user to log in.
FR-2.1	Profile	The system shall provide a form to input CGPA (0.0-10.0).
FR-2.2	Profile	The system shall allow the user to input the <i>number</i> of internships completed.
FR-2.3	Profile	The system shall allow the user to input the <i>number</i> of relevant certifications.
FR-3.1	Validation	The system shall validate that CGPA is a numeric value between 0.0 and 10.0.
FR-4.1	Prediction	The system shall provide a "Predict" button to authenticated users.
FR-4.2	Prediction	On request, the system shall calculate the composite API_Score.
FR-4.3	Prediction	On request, the system shall pass the full profile vector to the Random Forest model.
FR-5.1	Results	The system shall display the calculated API_Score (e.g., "7.8/10").
FR-5.2	Results	The system shall display the predicted path ("Placement," "Higher Studies," or "Startup").

Functional Requirements detailing the specific tasks the system must perform.

2.1.5. Other Non-Functional Requirements

Non-functional requirements define the quality attributes and constraints of the system.

Req. ID	Category	Requirement Description
NFR-1	Usability	The user interface must be simple, intuitive, and clutter-free.
NFR-2	Performance	The API score and career prediction must be generated and displayed in under 5 seconds .
NFR-3	Reliability	The system shall have 99.9% availability. The model must be periodically retrained to prevent drift.
NFR-4	Security	All user data must be transmitted over HTTPS. Passwords must be securely hashed.
NFR-5	Implementation	The backend must use Python, Pandas, and Scikit-learn. The frontend must use Streamlit.

Non-Functional Requirements specifying the quality attributes and performance criteria.

2.2. Data Flow Diagram

2.2.1. DFD Level 0

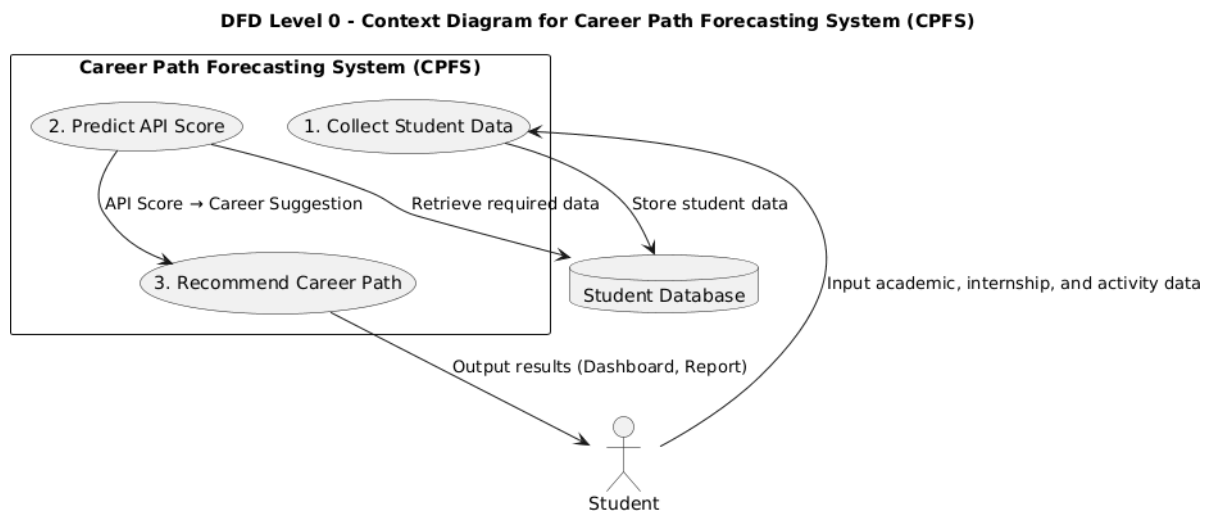


Figure 2.1: DFD Level 0

This diagram shows the system as a single process, interacting with the external entity (Student) through data flows.

2.2.2. DFD Level 1

This level decomposes the system into its major high-level processes.

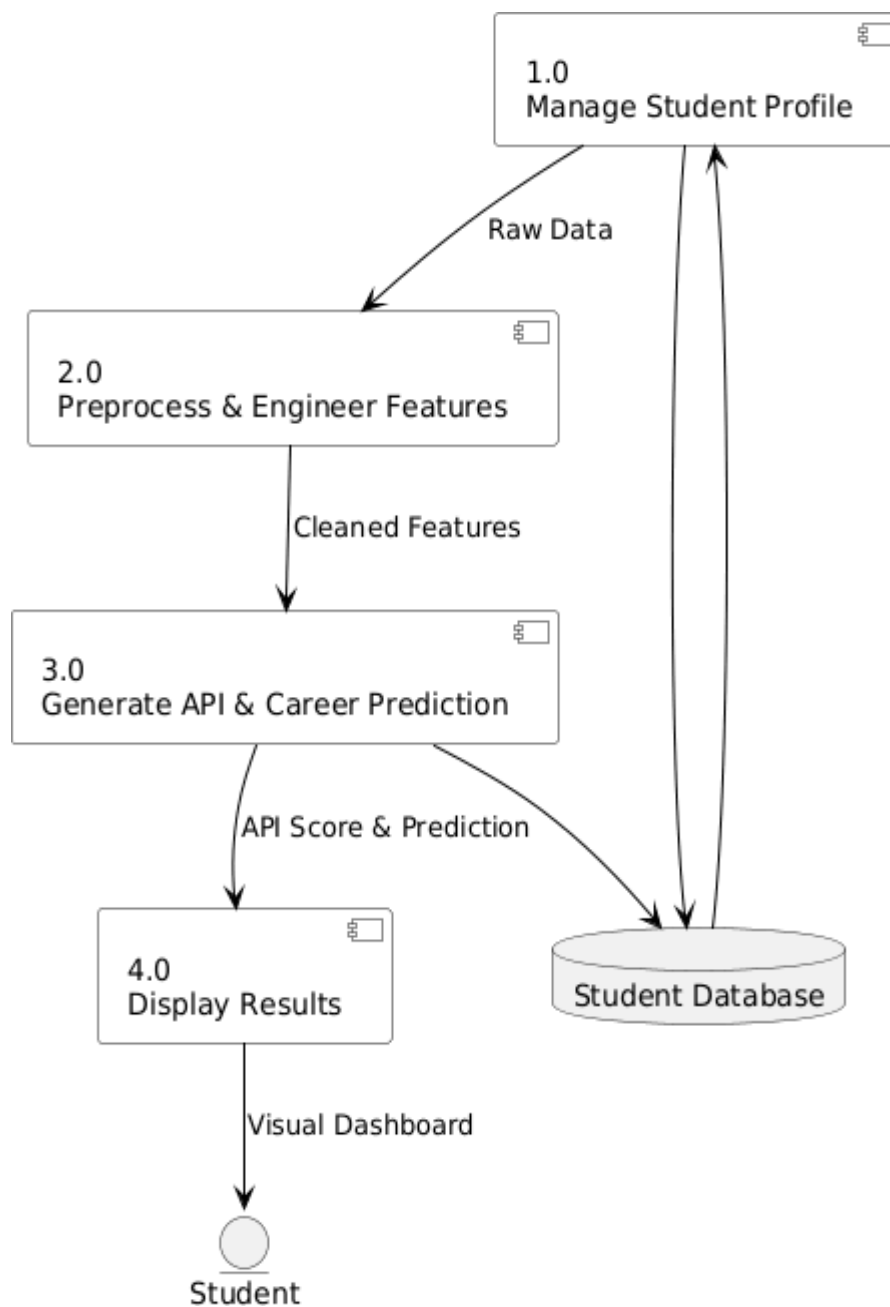


Figure 2.2: DFD Level 1

Level 1 DFD Showing Major System Processes and Data Flow

2.2.3. DFD Level 2

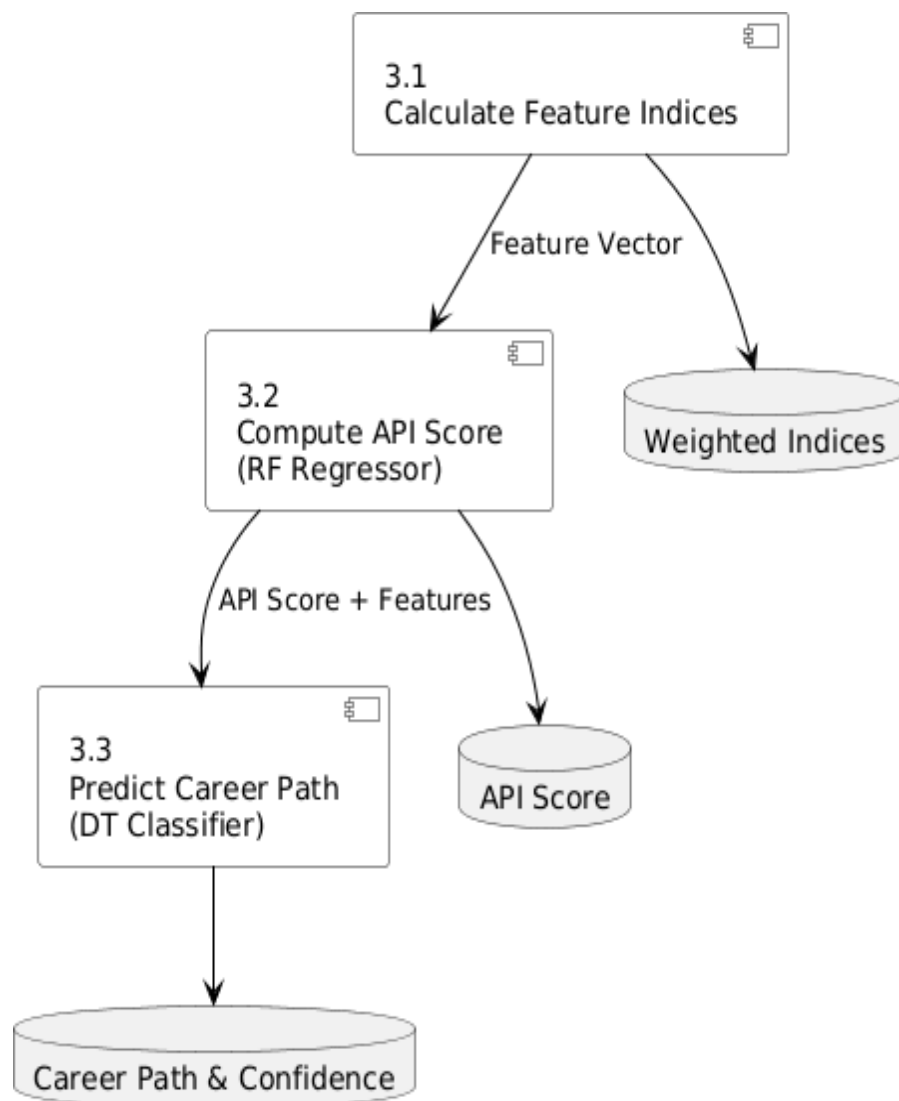


Figure 2.3: DFD Level 2

Level 2 DFD detailing the internal processes of the ML prediction engine.

2.2.4. Use Case Diagram

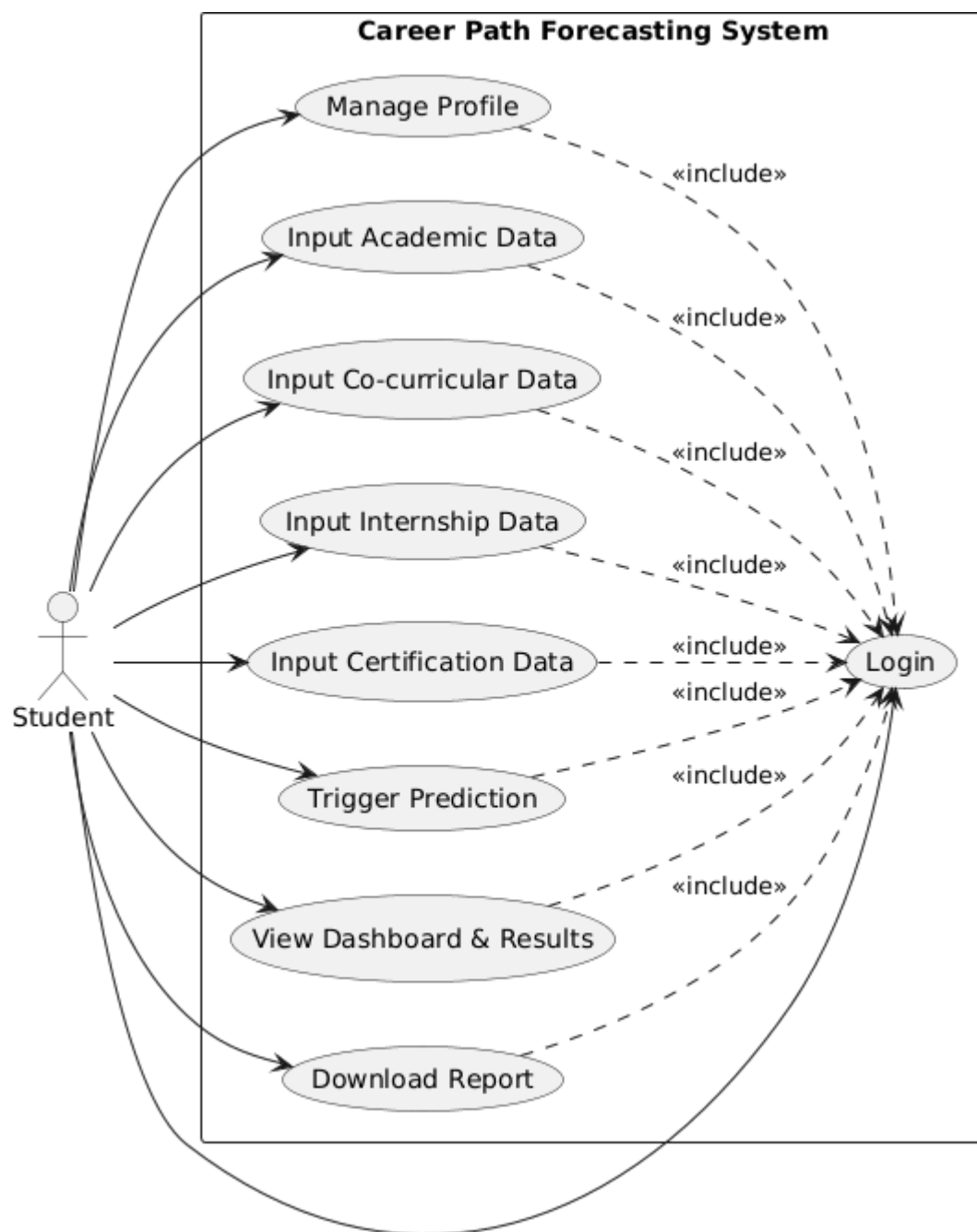


Figure 2.4: Use Case Diagram

Use Case Diagram illustrating all interactions between the Student actor and the system.

2.2.5. Activity Diagram

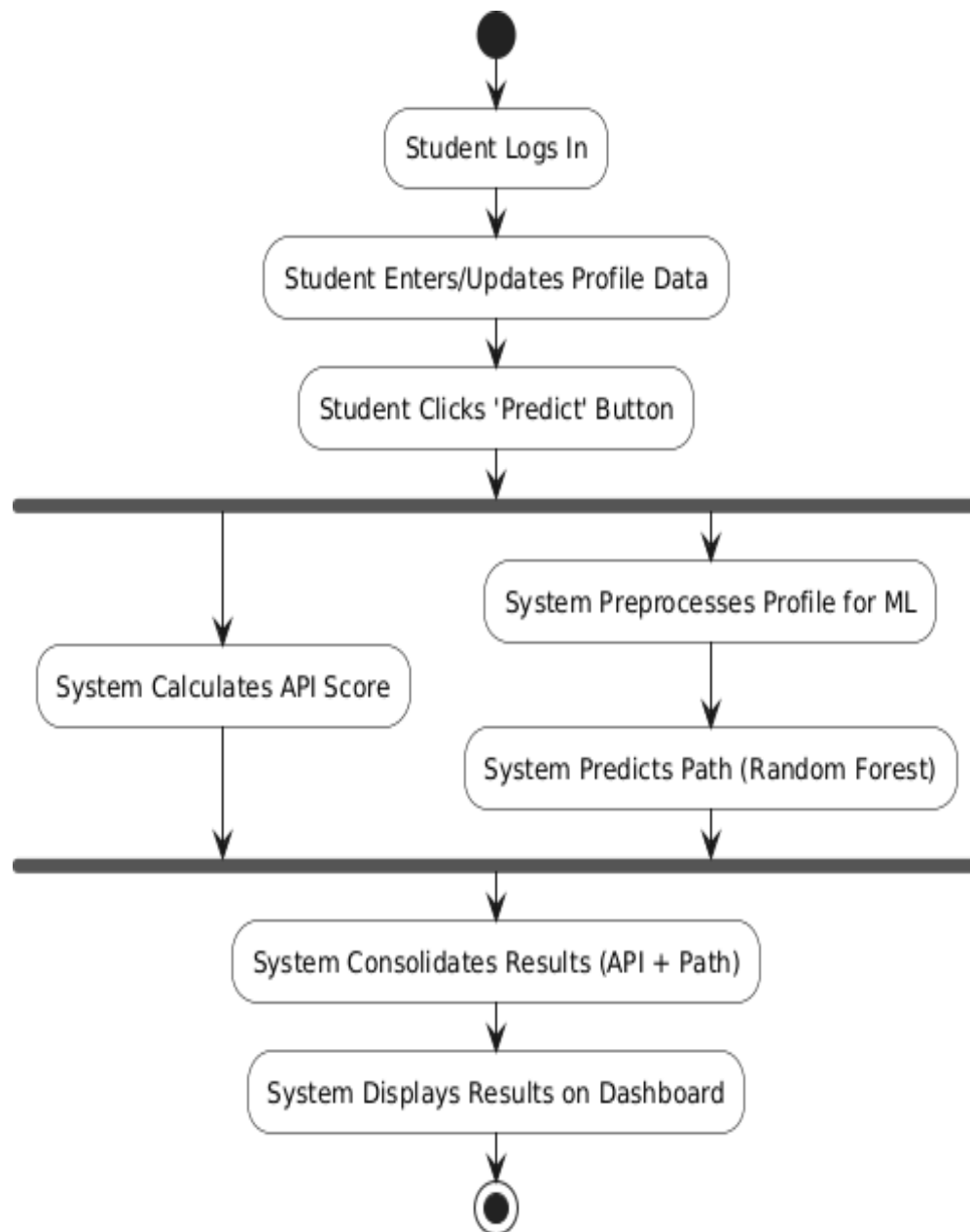


Figure 2.5: Activity Diagram

This diagram illustrates the control flow from the user's action to the final display of results, highlighting the sequential nature of the data processing and model execution.

2.2.6. State Transition Diagram (for a "Student Profile")

The state of a student's profile within the system transitions as follows:

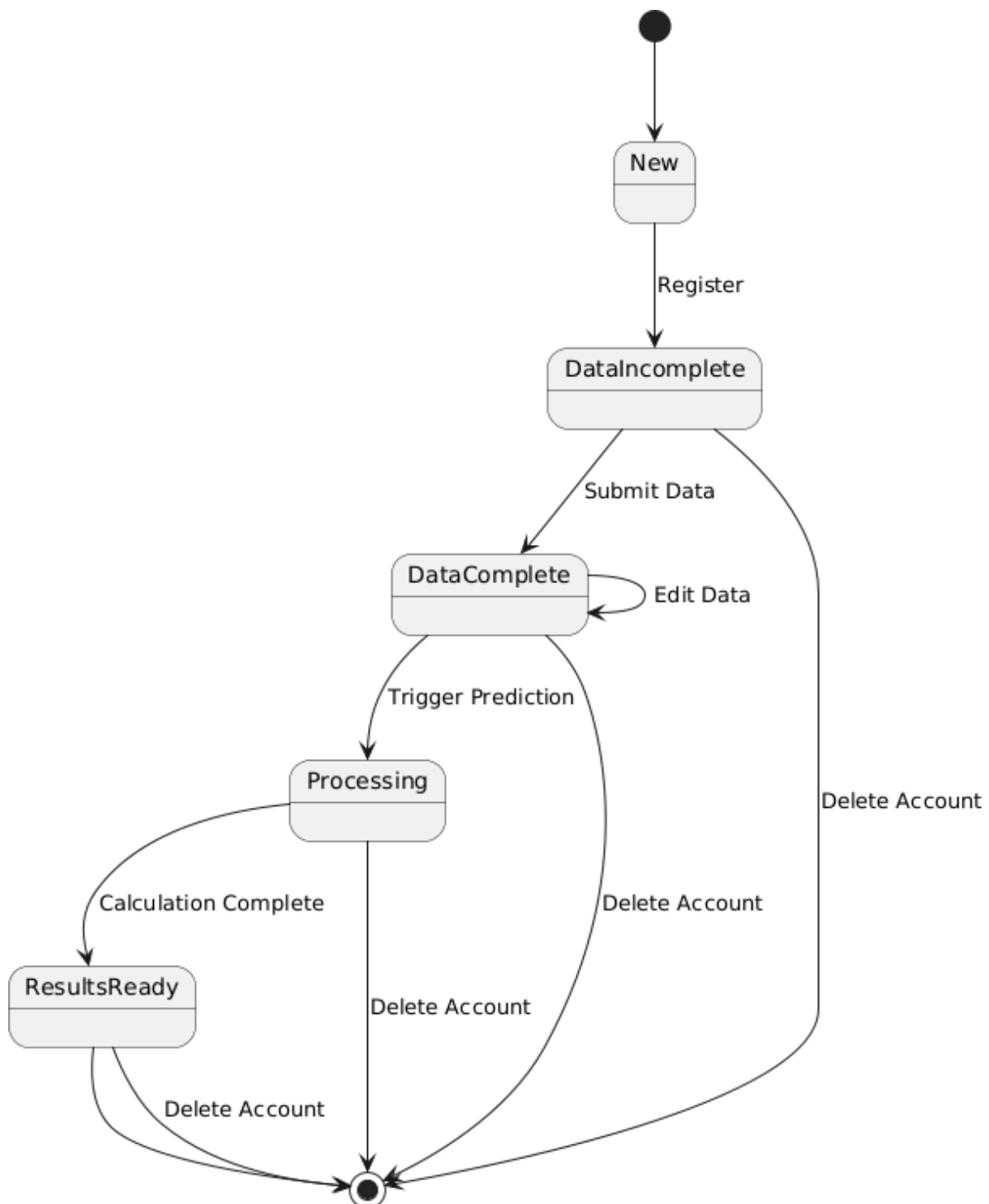


Figure 2.6: State Transition Diagram

State Transition Diagram showing the lifecycle states of a student profile.

2.2.7. Class Diagram

A detailed view of the system's static structure.

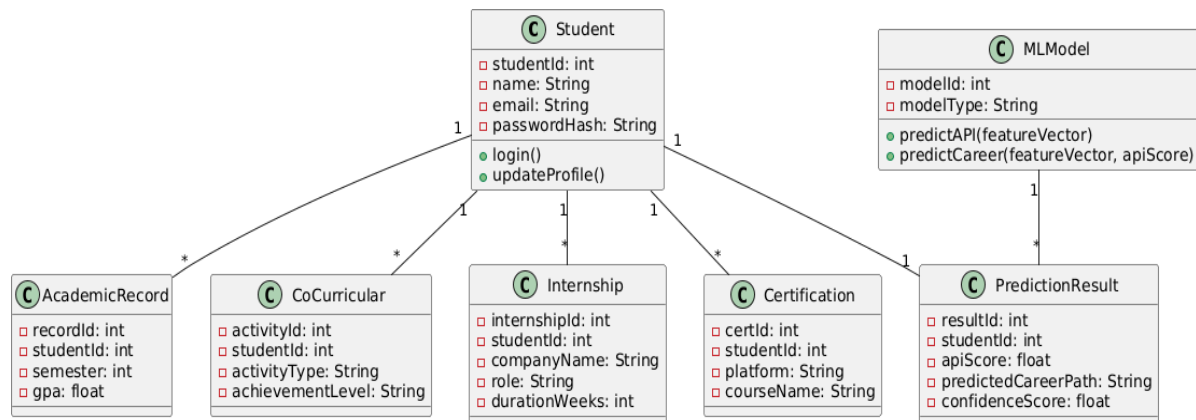


Figure 2.7: Class Diagram

Class Diagram defining the system's object structure and their relationships.

CHAPTER 3

PROJECT MANAGEMENT

3.1. Cost Estimation

3.1.1. Functional Point Estimation (FPE)

Function Point Analysis is a method to estimate software size based on its functionality.

Component	Description	Count	Weight (Avg)	Total
External Inputs (EI)	Login, Profile Form, Admin Upload	3	4	12
External Outputs (EO)	Results Dashboard, Error Messages	2	5	10
External Inquiries (EQ)	View Current Profile	1	4	4
Internal Logical Files (ILF)	User_DB, ML_Model	2	10	20
External Interface Files (EIF)	None	0	-	0
Total UFP				46

Characteristic (GSC)	Rating (0-5)
Data Communications	2
Distributed Data Processing	1
Performance	4
Heavily Used Configuration	3
Transaction Rate	3
On-Line Data Entry	5
End-User Efficiency	5
On-Line Update	4
Complex Processing (ML)	4
Reusability	3

Characteristic (GSC)	Rating (0-5)
Installation Ease	4
Operational Ease	4
Multiple Sites	1
Facilitate Change	3
Total GSC Score	46

Table 3.1: Project timeline breakdown showing phases, durations, and key deliverables

3.1.2. Efforts

Using an average productivity of **7 FP per Person-Month (PM)** for this project type:

- **Total Effort:** Total FP / Productivity = 51 FP / 7 FP/PM = **7.28 Person-Months**.
- **Team Size:** 4 members.
- **Schedule:** Effort / Team Size = 7.28 PM / 4 Persons = **1.82 Months**.
 - This timeline is consistent with a single-semester mini-project.

3.2. Risk Management

A risk management plan identifies potential problems and their mitigation strategies.

Risk ID	Risk Description	Prob.	Impact	Mitigation Strategy
R-01	Model Inaccuracy / Bias	M	H	Rigorous testing on a hold-out dataset; use a robust model (Random Forest); periodic retraining.
R-02	Data Privacy Breach	L	H	Enforce NFR-4: Hash passwords, use HTTPS, secure the database with access controls.
R-03	Insufficient Training Data	L	H	Mitigated: The model was successfully trained on a database of over 3000 students.
R-04	Model Drift	M	M	Schedule periodic retraining (e.g., annually) with new, recent student data to keep the model relevant.
R-05	Scope Creep	H	M	Strictly adhere to the signed-off SRS. Move all new feature requests to a "Version 2.0" backlog.

Risk ID	Risk Description	Prob.	Impact	Mitigation Strategy
R-06	Poor User Adoption	M	M	Conduct user acceptance tests (UAT) with peers to ensure the UI is intuitive and the results are trusted.

Table 3.2: Risk management matrix identifying potential risks and mitigation strategies.

3.3. Scheduling/Timeline

A Gantt chart provides a visual timeline. The project is broken down into 5 main phases over a 19-week period.

Phases & Milestones:

- **Phase 1: Planning & Requirements (Weeks 1-3)**
 - Finalize SRS Document (Milestone 1)
- **Phase 2: System & ML Design (Weeks 4-7)**
 - Complete Architectural & Database Design
 - Finalize ML Model Architecture & Data Pipeline (Milestone 2)
- **Phase 3: Implementation (Weeks 8-13)**
 - Backend Development (Data Processing, API)
 - Frontend Development (Streamlit UI)
 - ML Model Training & Validation (Milestone 3)
- **Phase 4: Testing & Integration (Weeks 14-16)**
 - Unit, Integration, and System Testing
 - User Acceptance Testing (UAT) & Bug Fixing (Milestone 4)
- **Phase 5: Deployment & Documentation (Weeks 17-19)**
 - Deploy to Production Server
 - Finalize Project Report & Documentation (Milestone 5)

CHAPTER 4

DESIGN ENGINEERING

4.1. Architectural Design

The system uses a **3-Layer Architecture** (a variant of Model-View-Controller) adapted for a data-science application.

1. **Layer 1: Presentation Layer (View)**

- **Technology: Streamlit**
- **Responsibility:** Renders the entire user interface, captures user input, and displays system output.

2. **Layer 2: Business Logic Layer (Controller)**

- **Technology: Python, Pandas**
- **Responsibility:** Acts as the "brain." It handles user authentication, data validation, and the algorithm for **calculating the API Score**.

3. **Layer 3: Data & Model Layer (Model)**

- **Technology: Scikit-learn, Database (SQLite), File System**
- **Responsibility:** Manages the User_DB for data persistence and the pre-trained random_forest.pkl model for prediction.

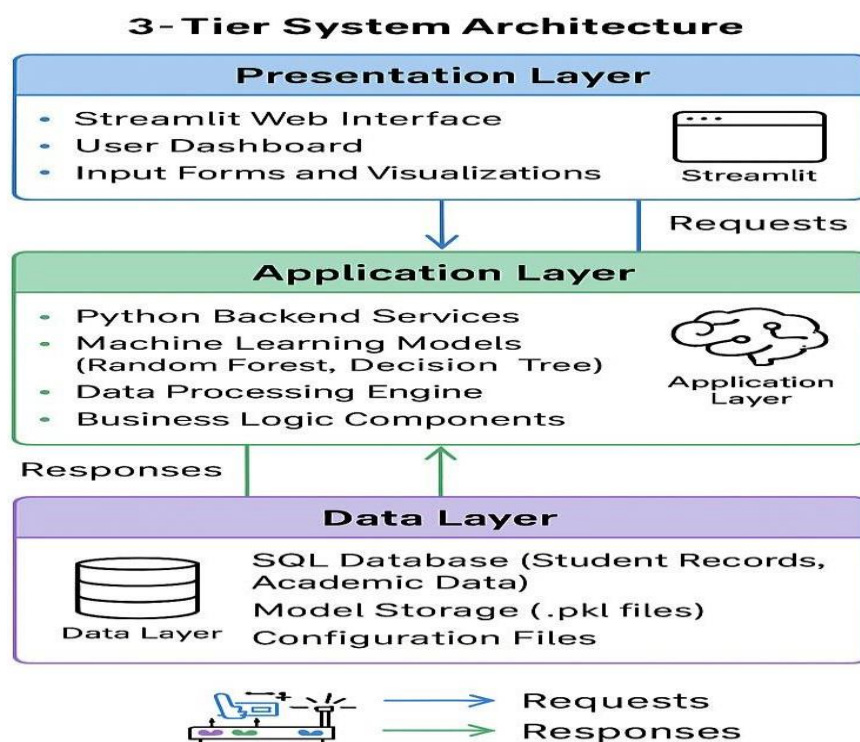


Figure 4.1:

3-Tier Architecture Diagram showing the Presentation, Application, and Data layers with their respective components and data flow.

4.2. Data design

The data design is based on the ERD and implemented in a relational database.

Field Name	Data Type	Constraints	Description
student_id	INT	PRIMARY KEY	Unique student identifier
email	VARCHAR (255)	UNIQUE, NOT NULL	User's login email
password_hash	VARCHAR (255)	NOT NULL	Hashed and salted password
student_name	VARCHAR (100)		User's full name

Field Name	Data Type	Constraints	Description
profile_id	INT	PRIMARY KEY	Unique profile identifier
student_id	INT	FOREIGN KEY	Links to Students table
cgpa	DECIMAL (4, 2)	NOT NULL	User's CGPA
internship_count	INT	NOT NULL, DEFAULT 0	Number of internships
certification_count	INT	NOT NULL, DEFAULT 0	Number of certifications
cocurricular_score	INT	NOT NULL, DEFAULT 0	Score for activities
last_updated	TIMESTAMP		When the profile was last edited

Table 1: Component Specification detailing the functionality and dependencies of each module.

Table 2: API Score Calculation Metrics showing the weighted parameters for holistic evaluation.

4.3. Component Level Design

The component design describes the internal structure of key software modules.

- **app.py (Streamlit):** This component handles all UI rendering. It uses Streamlit's session_state to manage user authentication. It contains the logic to render forms (st.slider, st.number_input) and display results (st.success, st.metric). On a button click, it calls the PredictionEngine.
- **PredictionEngine.py (Class):** This class encapsulates all backend logic.
 - `__init__()`: Loads the random_forest.pkl model into memory on startup.
 - `calculate_api(data)`: Contains the business logic to calculate the 0-10 API score from the profile data.
 - `get_prediction(data)`: Preprocesses the input data into the exact vector the model expects, calls `self.model.predict()`, and post-processes the numeric output (e.g., 0) into a string (e.g., "Placement").

CHAPTER 5

TESTING

5.1. Model Evaluation (ML Testing)

- **Objective:** To ensure the Random Forest model is accurate, reliable, and unbiased.
- **Method:** The model was trained on a 70-30 split of the 3000+ student dataset.
- **Metric:** The primary metric was **Mean Absolute Error (MAE)** for API score regression and **Accuracy/F1-Score** for career path classification.
- **Validation:** **K-fold Cross-Validation (k=10)** was used during training. This ensures the model's performance is stable and not a result of a lucky split.

5.2. Software Testing (Test Case Examples)

Unit Testing

Testing individual functions in isolation.

Test ID	Component	Function/Test	Test Description	Expected Result
UT-01	PredictionEngine	calculate_api()	Input a profile with maximum values.	Returns 10.0.
UT-02	PredictionEngine	calculate_api()	Input a profile with minimum (zero) values.	Returns 0.0.
UT-03	app.py	(Validation)	Enter "abc" into the CGPA field.	An error message is displayed.

Table 1: Test Case Summary showing pass/fail results for all functional requirements.

Integration Testing

Testing the flow of data between components.

Test ID	Components Tested	Test Description	Expected Result
IT-01	app.py -> PredictionEngine	User clicks "Predict".	get_prediction is called, and render_results displays the output.
IT-02	app.py -> UserManagement	User clicks "Register".	A new record is created in the Students table.

Table 2: Model Evaluation Metrics presenting accuracy, precision, and recall scores for ML models.

System Testing

Testing the complete, deployed application as a user.

Test ID	Use Case	Test Description	Expected Result
ST-01	UC-01 (Happy Path)	1. Register. 2. Login. 3. Fill profile. 4. Click Predict.	The user is successfully created, logged in, and sees a valid recommendation.
ST-02	NFR-2 (Performance)	Time from clicking "Predict" to results appearing.	The duration is < 5 seconds.
ST-03	NFR-4 (Security)	Attempt to access the profile page URL without logging in.	The user is redirected to the login page.

Table 3: Performance Test Results showing response times under different user loads.

CHAPTER 6

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